

SFU Beedie Business Analytics Hackathon 2025

BC Hydro's Supply Chain

Risks, Insights, and Solutions for Efficiency

Team 9 - Div.2

Isabel Angwah
Sahiljeet Singh
Jane (Trang Le)



Overview

- Business Context & Challenges
- Analysis: Our Methodologies
- Risk Impact on Business
- Potential Solutions
- Limitations

Organization Context

- BC Hydro is a Crown corporation supplying electricity across B.C.
- Relies on global and regional vendors for infrastructure needs.
- Procurement spend has tripled due to infrastructure growth:

  Total Annual Spend: $599,258,366

Organization Snapshot

- Contracts expire in waves from 2025–2032.
- High spend on switchgear, construction, fleet, and components.
- Relies heavily on vendors in Canada, U.S.
- Some items face 900+ day lead times and low stock.



Key Challenges in BC Hydro's Supply Chain

Supply Chain Risk Exposure & Emerging Threats

- Tariff Exposure
- Vendor Concentration & Geographic Risk
- Lead Time and Inventory Gaps
- Contract Expiry Cliff

Methodologies

Risk Assessment

- Identify supply chain risks
- Prioritize risks by category
- Visualize risk heatmap

Tariff Simulation

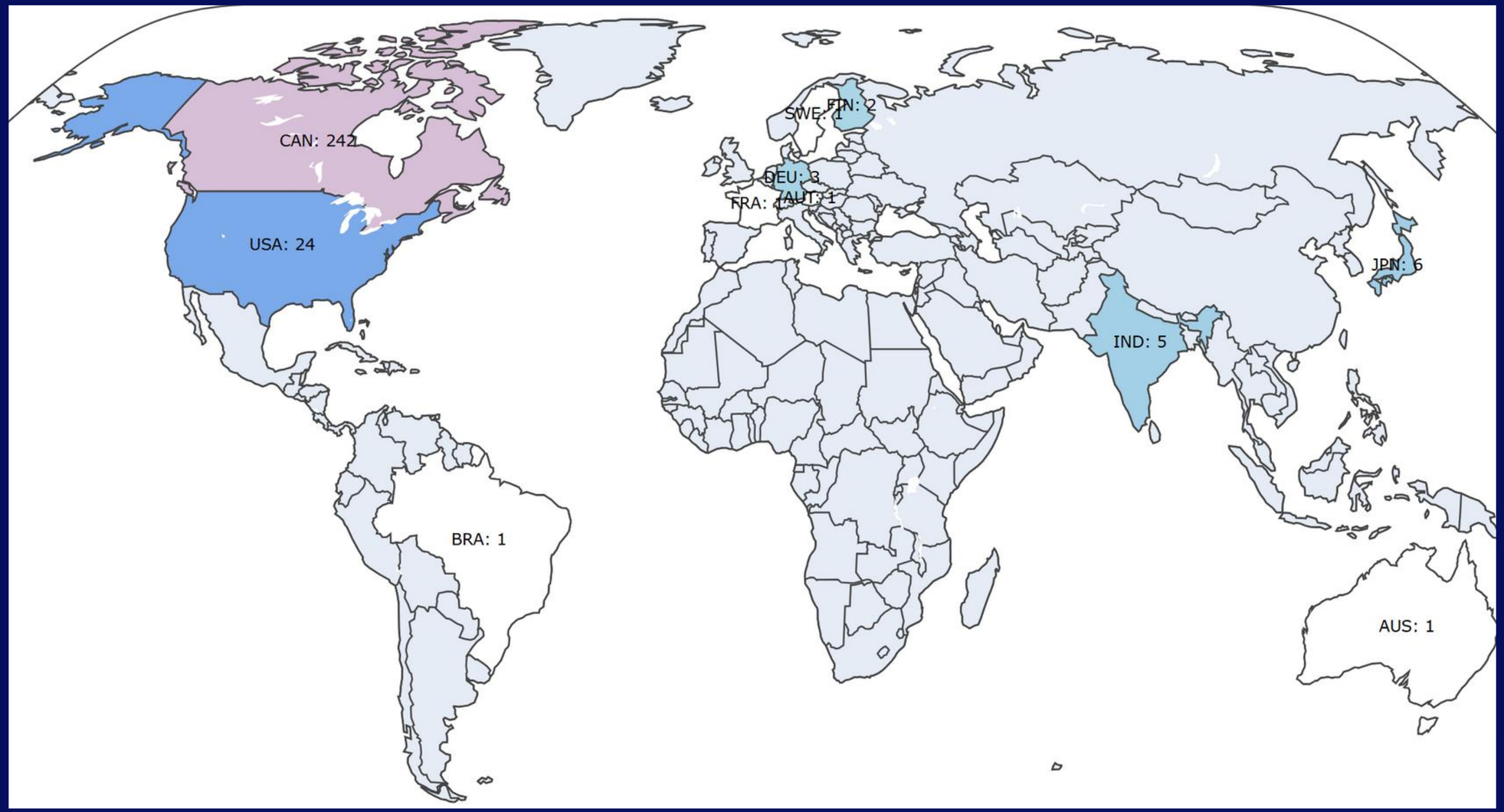
- Model tariff impact on costs
- Determine exposure of categories
- Simulate 25% tariff scenario

Supplier Evaluation

- Analyze alternative suppliers
- Assess availabilifty & cost
- Find opportunities to diversify

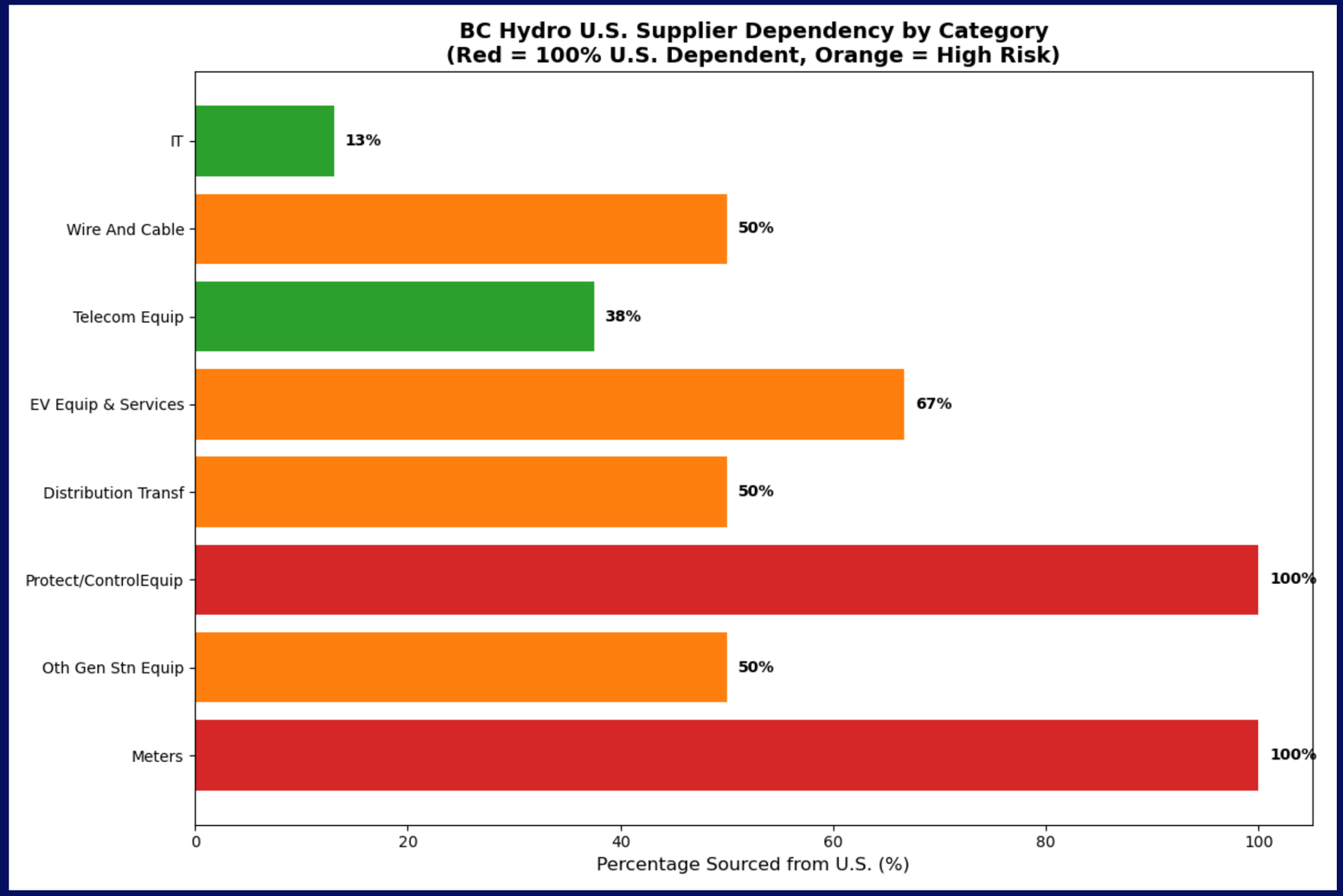


Risk Exposure:
Vendor Analysis



Category Based Risk Assessment

- Total Equipment Categories Analyzed: 15
- Total Annual Spend: $599.3 million
- U.S.-Sourced Spend: $202 million
- High-Risk Categories (50%+ U.S.
- sourced): 6



Risk Insight

- Red bars in the chart = 100% U.S. dependent — extremely vulnerable
- Orange bars = 50–67% U.S. dependence — still high risk

Supply Chain Risk vs. Spending Impact

- **Meters & Protect/Control Equip**

→ High risk, 100% U.S. dependency — critical exposure.

- **Other General Station Equipment & Distribution Transformers**

→ High spend, 50% U.S. dependency — cost-heavy and vulnerable.

- **EV Equipment**

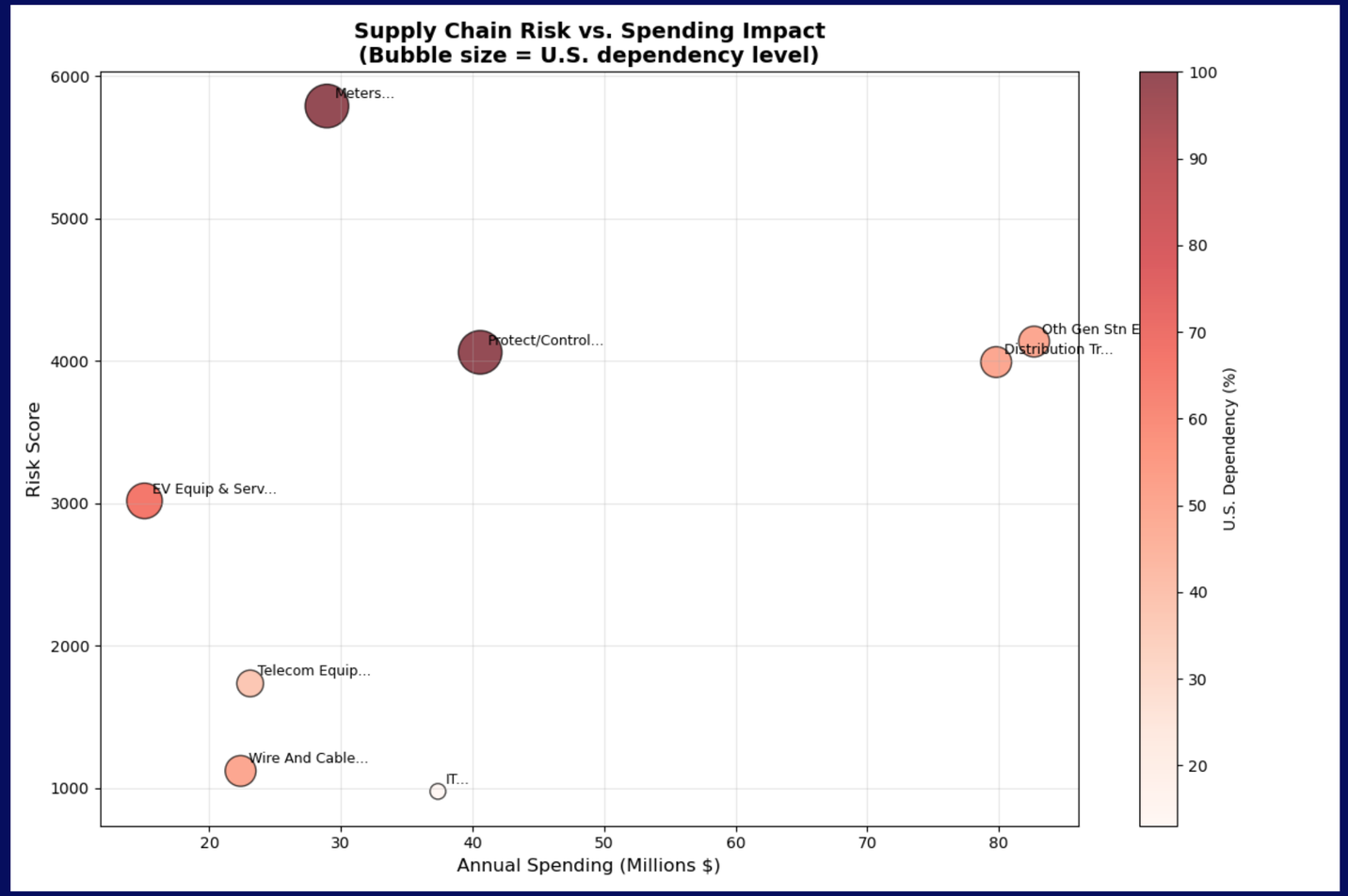
→ Mid spend, high risk, 67% U.S. dependent: under-the-radar risk.

- **IT**

→ High spend, low risk, minimal U.S. dependency - strategically safe.

- **Telecom & Wire/Cable**

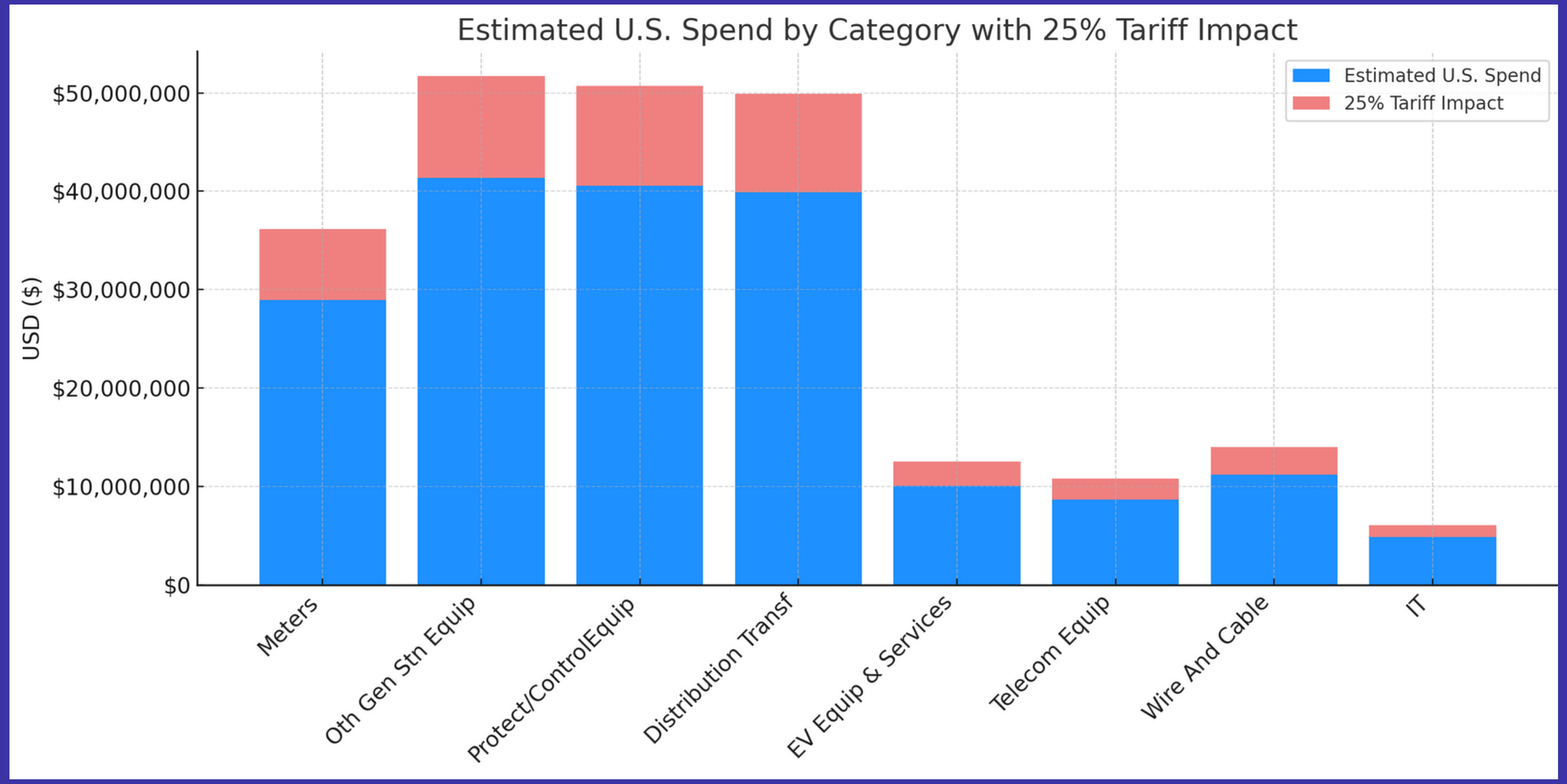
→ Low risk, low spend, moderate exposure — manageable.



Impact on BC Hydro

What our findings mean...

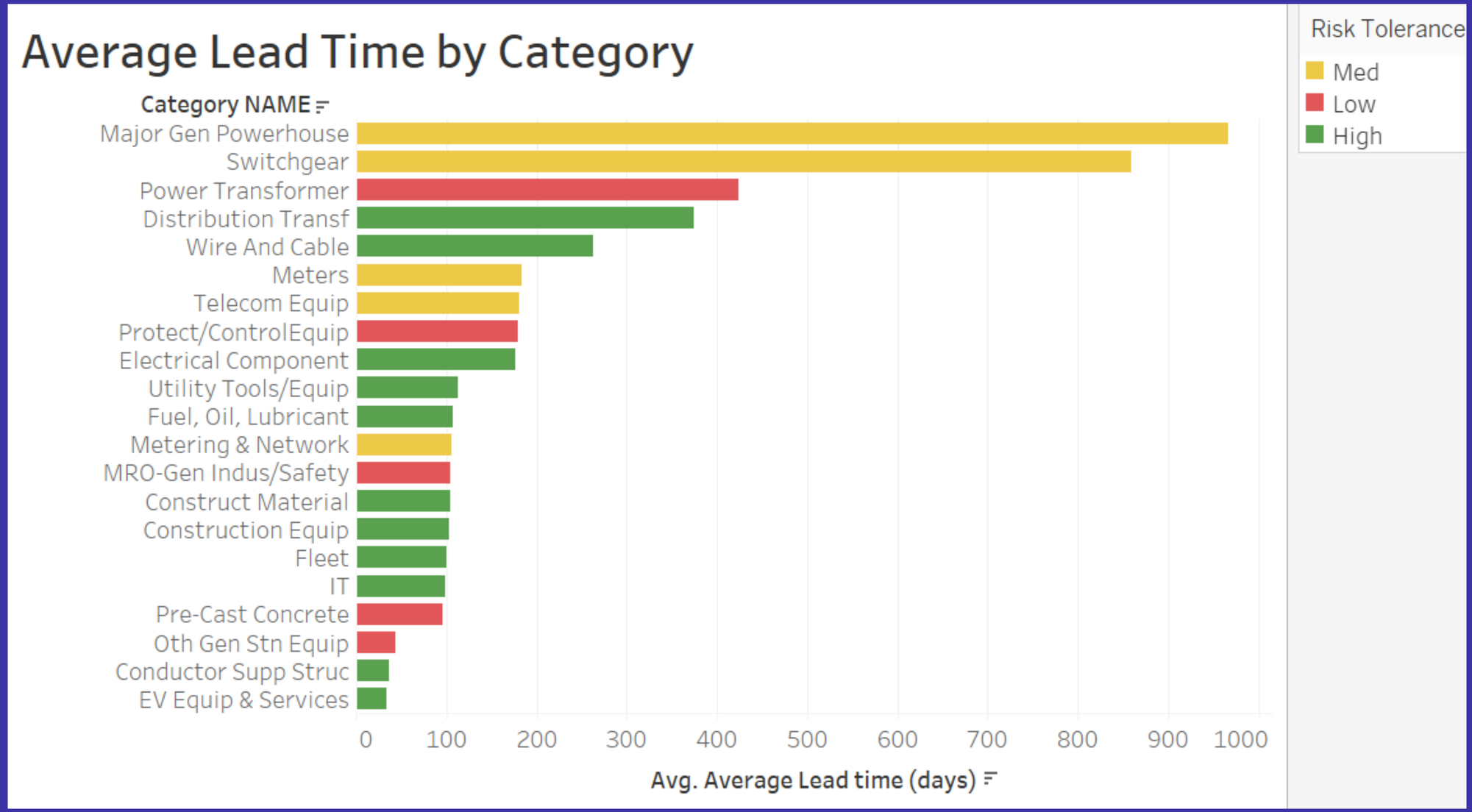
- Financial Risk:
  - Retaliatory trade tariffs on U.S. suppliers could increase costs by millions for meters, protect/control equipment.



Impact on BC Hydro

What our findings mean...

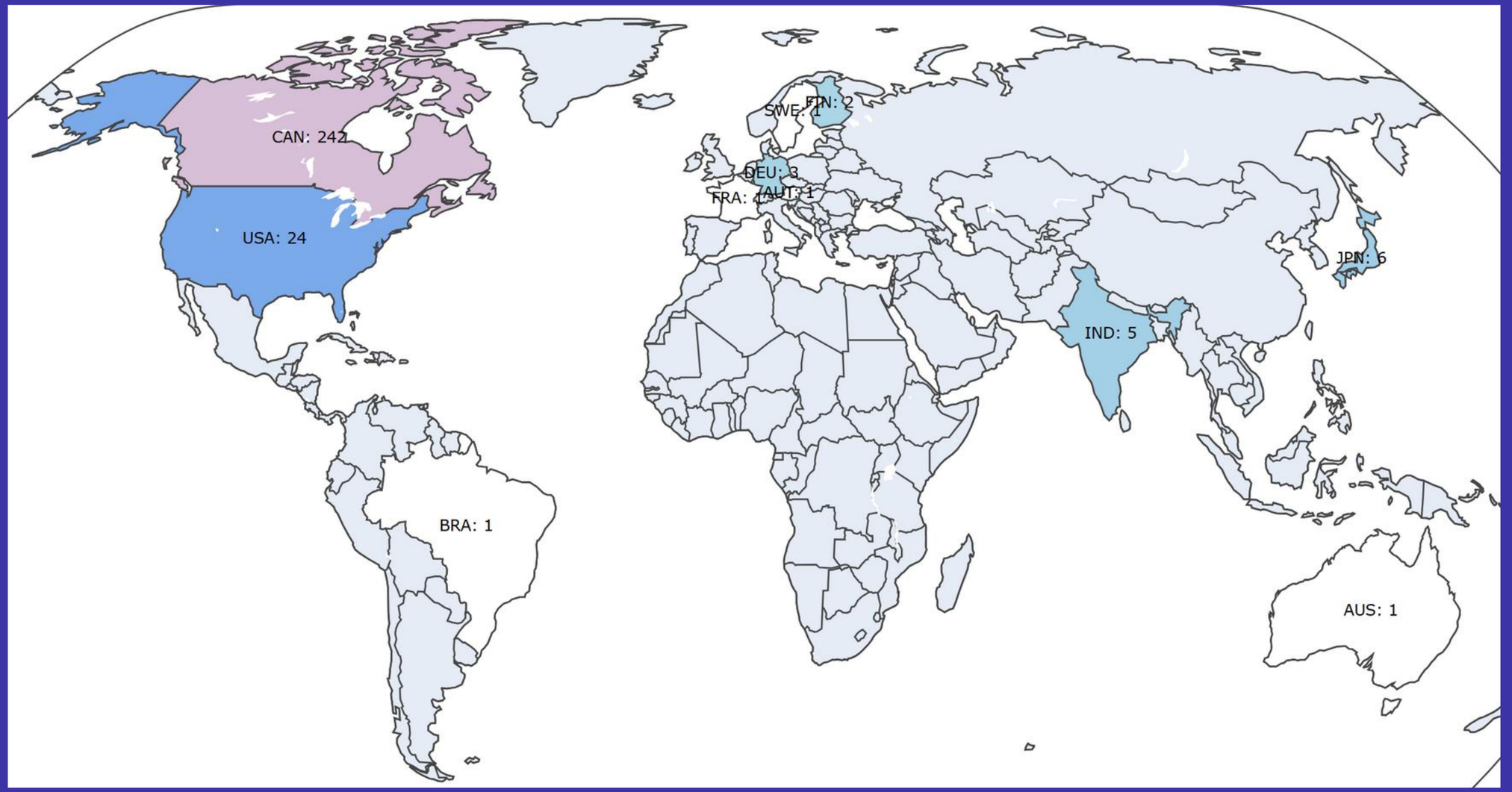
- Operational Risk:
  - Unstable and slow delivery times may delay important power projects.
    One delay in getting a transformer = power outages



Impact on BC Hydro

What our findings mean...

- Strategic Risk:
  - Over-reliance on select vendors and countries weakens supply chain resilience and service continuity



Potential Solutions

Short-Term

0-6 months

- Order early & stock up on US-dependent parts to avoid delays

→ Keeps projects on schedule

- Launch real-time risk dashboard

→ Enables faster, informed decisions



Medium-Term

6-18 months

- Diversify suppliers beyond U.S. & Canada

→ Reduces tariff risk, adds flexibility, mitigate geopolitical issues

- Strengthen vendor agreements

→ Improves delivery reliability



Long-Term

> 18 months

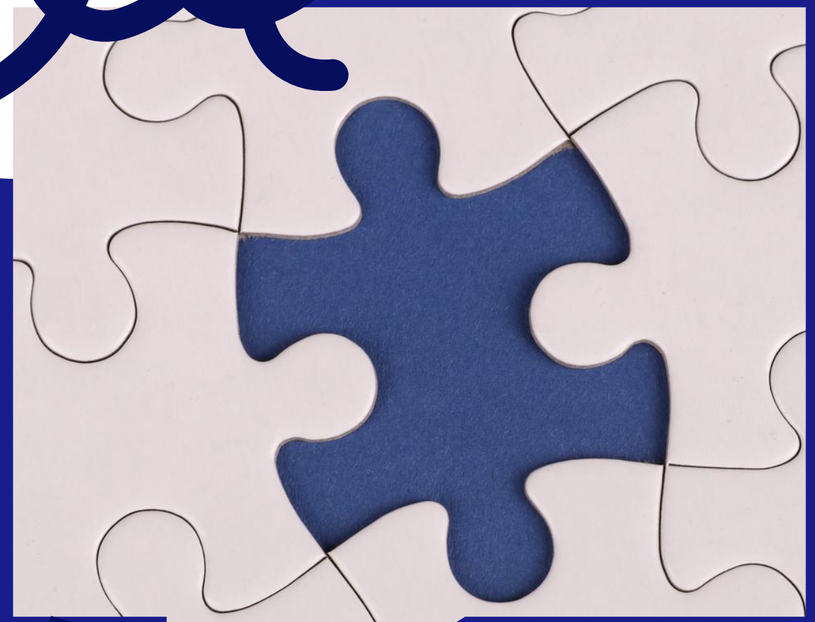
- Add risk scoring to procurement

→ Builds long-term supply resilience



Limitations

- **Stocking on US restricted products:**
  Adds to existing costs and needs better inventory management
- **New vendors:**
  Not always available or reliable
- **Data Gaps and Inconsistent Inputs from Vendors**





Thank you!